Joshua Trujillo

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EDUCATION

University of Texas at Austin | Austin, TX ***GPA: 3.8*** **| May 2027**

Mechanical Engineering

- Technical Skills: SolidWorks, AutoCAD, GD&T, Revit, Python, MATLAB, C++, Arduino, Finite Element Analysis (FEA), 3D printing, Machine shop, Embedded systems, Soldering, Excel
- Relevant Schoolwork: Heat Transfer, Fluid Mechanics, Dynamic System and Controls, Mechatronics

EXPERIENCE

Longhorn Boring | Austin, TX **Sep 2026 – Present**

Mechanical Engineering team

- Evaluate 10+ tunnel boring machine concepts based on excavation performance, manufacturability, cost, and competition constraints to support mechanical design selection.
- Develop and compare mechanical concepts in SolidWorks, considering component geometry, packaging, structural requirements, and subsystem integration.

Walker Department of Mechanical Engineering |Austin, TX **May 2026 – Present**

Research Assistant | ***Advisor: Prof. Ofodike Ezekoye***

- Designed and fabricated an ASTM E681 flammability apparatus, applying heat-transfer calculations and published research to size components and automate temperature control for testing.
- Designed and validated a low-speed wind tunnel for fire and thermal research, using CFD, Type K thermocouples, and Schmidt-Boelter heat-flux gauges to assess airflow and thermal conditions; reduced measurement error from approximately 55% to under 10%.
- Instrumented large-scale LFP and NMC battery thermal-runaway tests with 200+ Type K thermocouples, eight Directional Flame Thermometers, load cells, and air-quality sensors.
- Collected and analyzed temperature, heat flux, gas, smoke, particulate, load, video, and acoustic data using Graphtec software to evaluate thermal propagation and fire behavior.
- Supported safe execution of high-hazard testing by establishing baseline air measurements, verifying instrumentation, coordinating test procedures, and following PPE and fire-safety protocols involving respirators, protective suits, and extinguishers.

Hanger Inc. | Austin, TX **Jun 2024 – Jul 2024**

Finance and Accounting Intern

- Supported the finance and accounting team in analyzing cash flows and ensuring accurate financial reporting across company operations
- Assisted in the Unapplied Cash Project, reconciling payment discrepancies of ~$30 million to improve cash application efficiency
- Conducted Telecom ID matching to streamline expense tracking and verify cost allocations saving ~$10,000 in unnecessary spending

PROJECTS/ACTIVITIES

Automated Irrigation System | Austin, TX **Jan 2026 – Present**

Personal Project

- Designed and implemented an IoT-controlled irrigation system using Raspberry Pi and Python to automate a DC water pump
- Integrated a solar-powered renewable energy system to enhance sustainability and energy efficiency
- Integrated humidity, temperature, and water-level sensors for real-time environmental monitoring and automated pump control

1 lb Battle Bot Project | Austin, TX **Jan 2025 – May 2025**

Design and Manufacturing team

- Designed and modeled a 1 lb combat robot using SolidWorks, adhering to SPARC competition guidelines
- Contributed to mechanical design, assembly, and performance testing
- Placed 3rd out of 20 teams in a competitive robotics tournament

Club Soccer | Austin, TX **Aug 2023 – Present**

Member/Captain

- Led 35+ members to 3 national appearances scoring and assisting 48 & 34 goals respectively